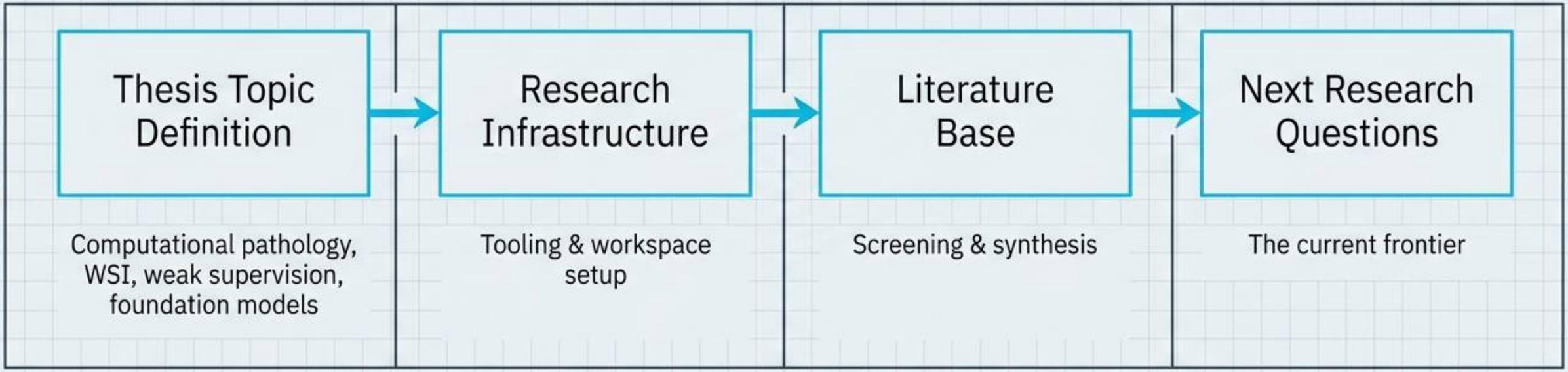

The first 14 days focused on orientation and workflow stabilization. Goal: Build a systematic map of computational pathology to support targeted experimental design, ensuring a reproducible and automated research environment from day one.

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Structuring the Digital Workspace for Scalable Research

Workspace Anatomy Map

Active Workspace

`papers/`

`docs/`

Primary zones for literature screening, thematic synthesis, and foundational thesis notes.

Automation

`.agents/skills/bib/`

`.codex/skills/pdf/`

Project-local and user-level skills to reduce reduce manual overhead.

Inherited Baseline

INHERITED

`code/TCGA-UT/`

Copied legacy code from prior university employee. Held purely as context for orientation and future baseline analysis, not original implementation.

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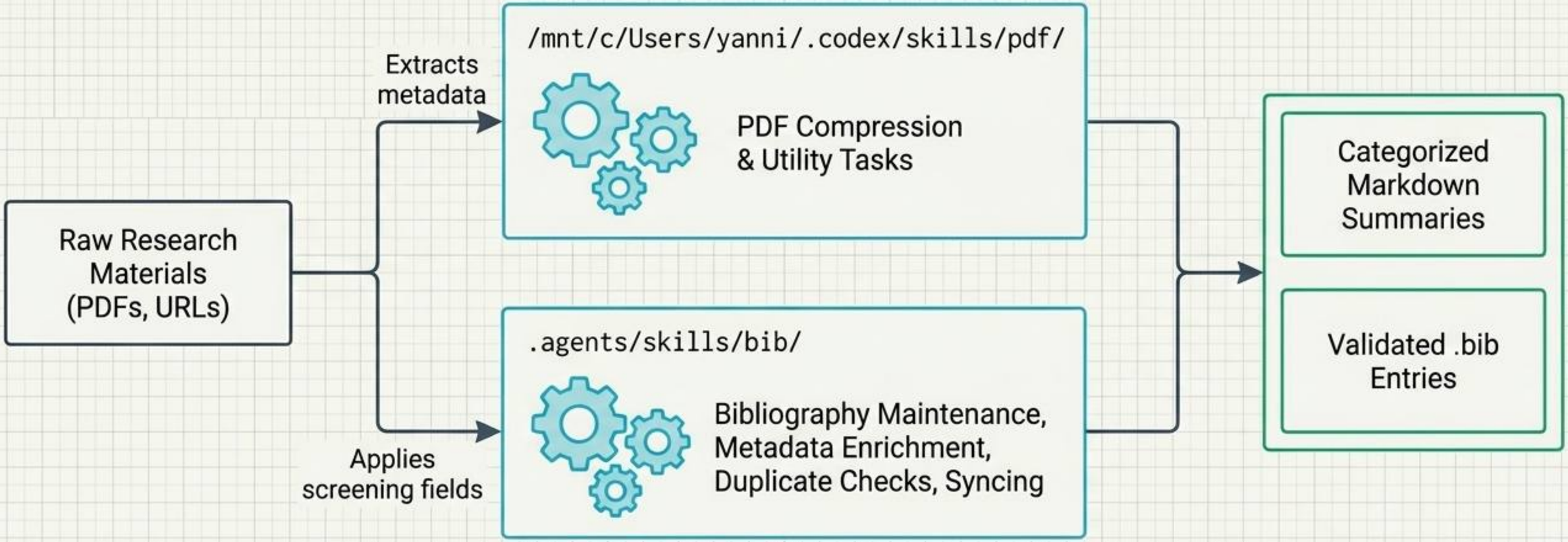
Automating Repetitive Workflows with Agent Skills



Custom tooling built to eliminate administrative friction during continuous literature review.

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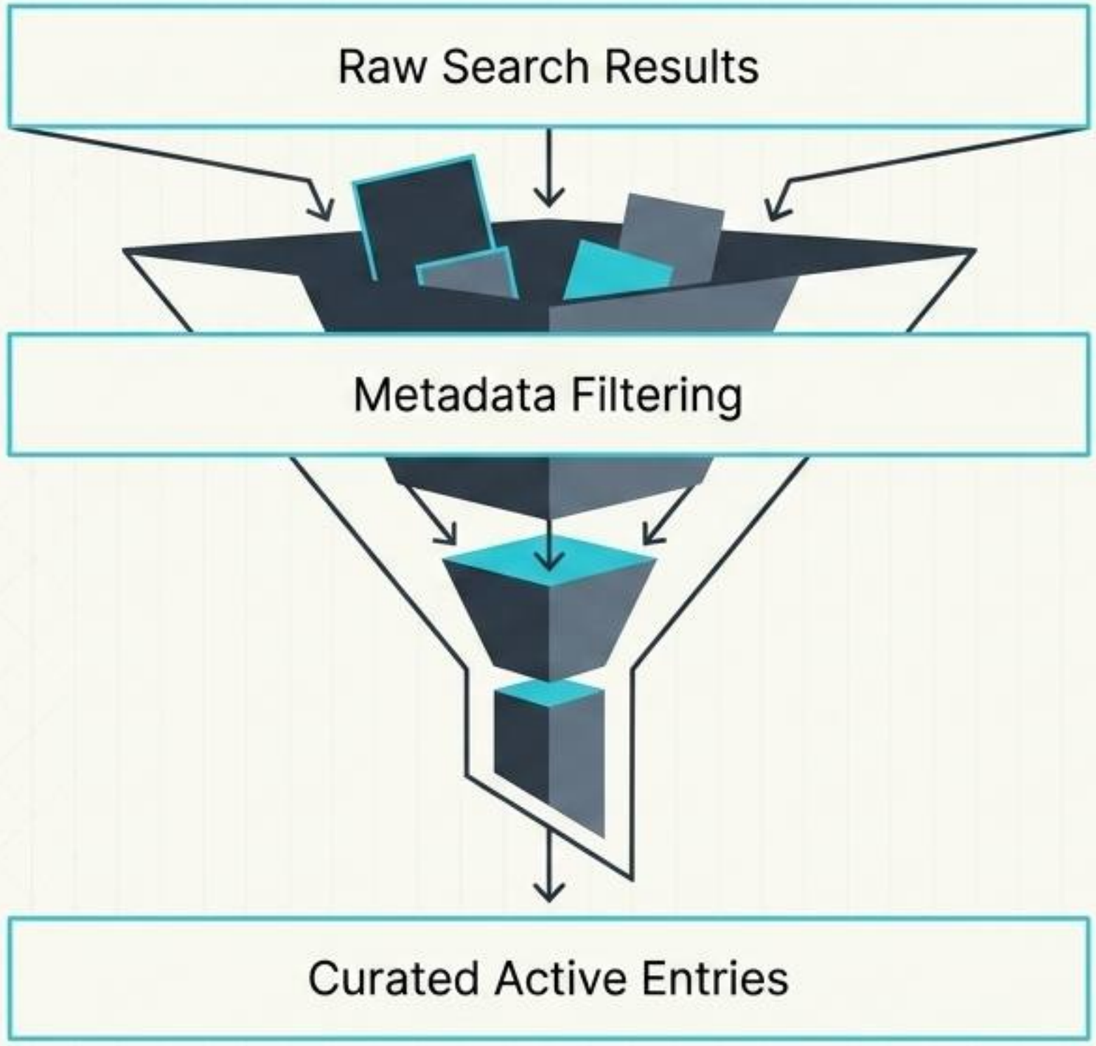
Compiling and Screening the Foundational Literature



Count Matrix

27

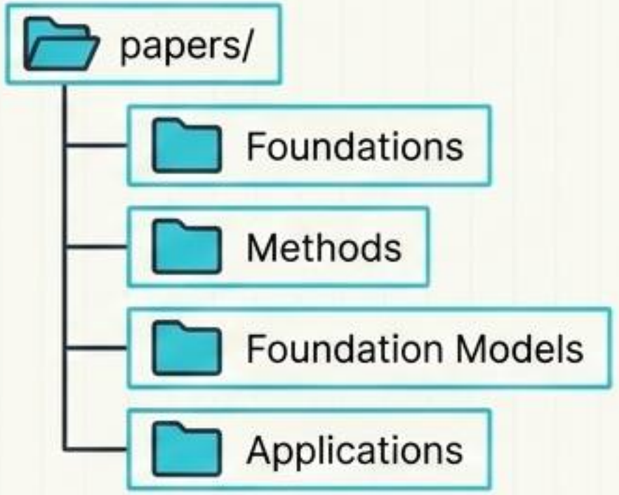
Active Sources

`mapped to papers/sources.bib`

4

Excluded / Separate Review

`mapped to papers/exclusions.bib`



Active sources are distributed into thematic folders, making the growing library highly inspectable and update-friendly.

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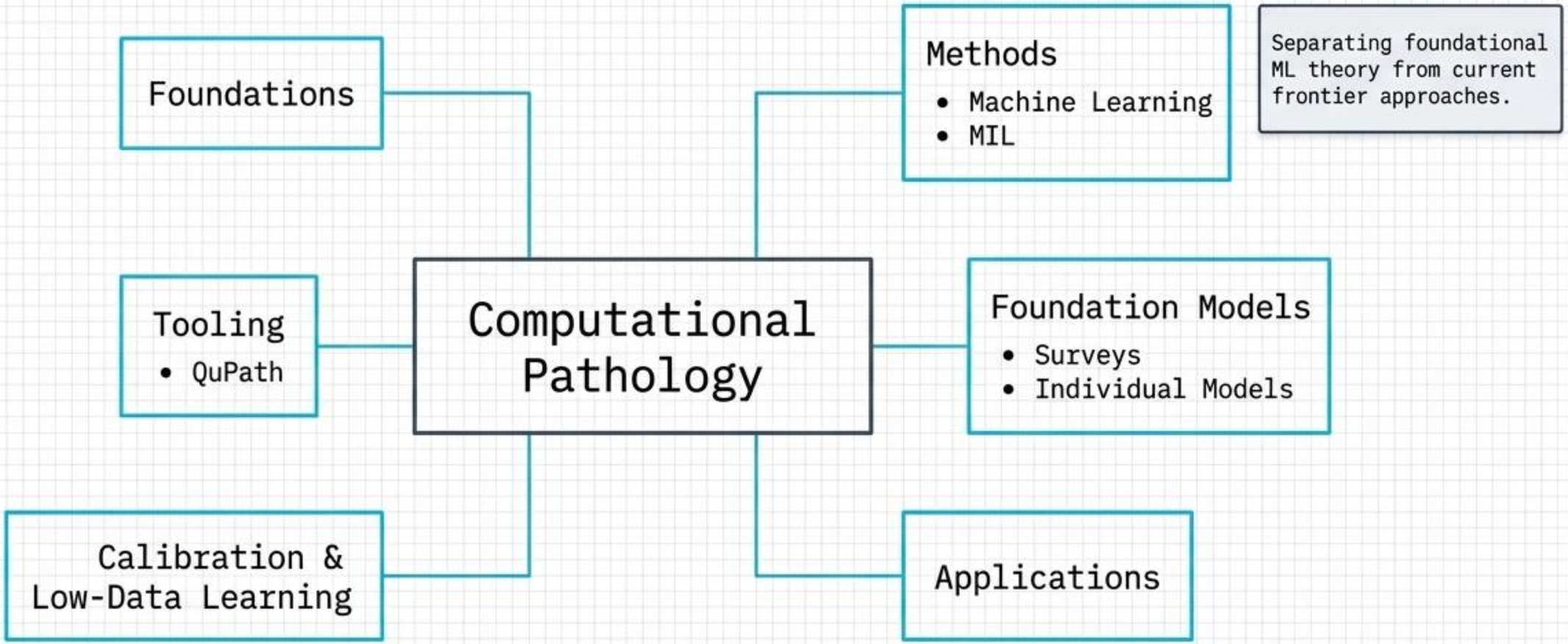
Mapping the Thematic Universe of Computational Pathology



Takeaway: This thematic clustering actively separates established background context from cutting-edge model research.

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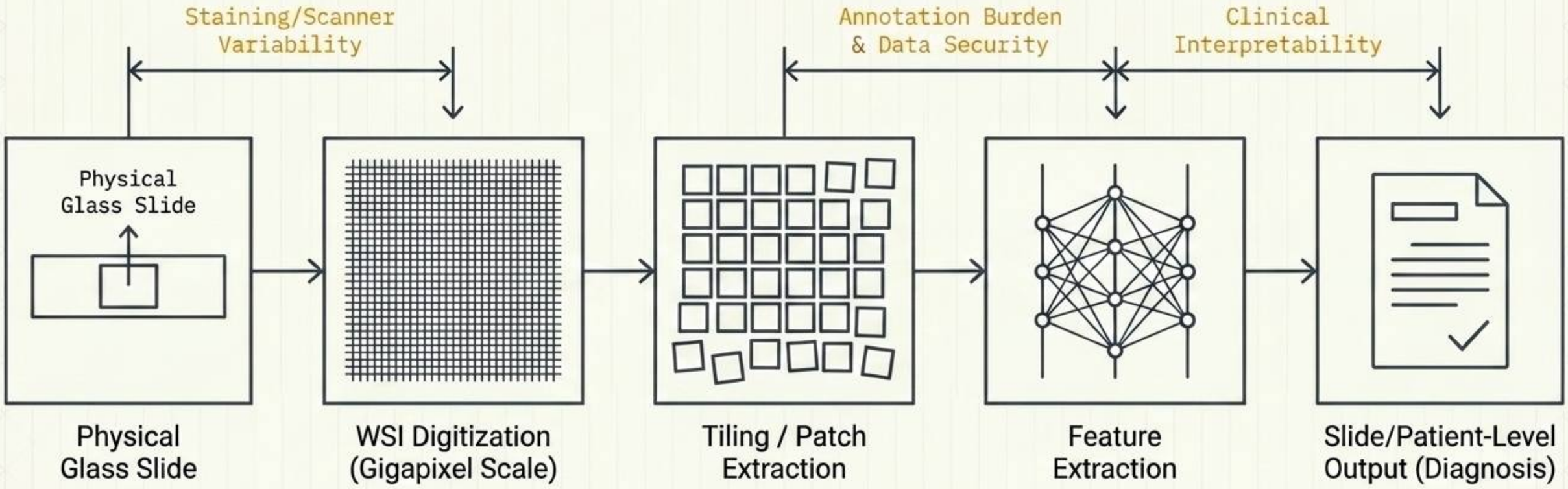
Distilling the Computational Pathology Dgitization Pipeline

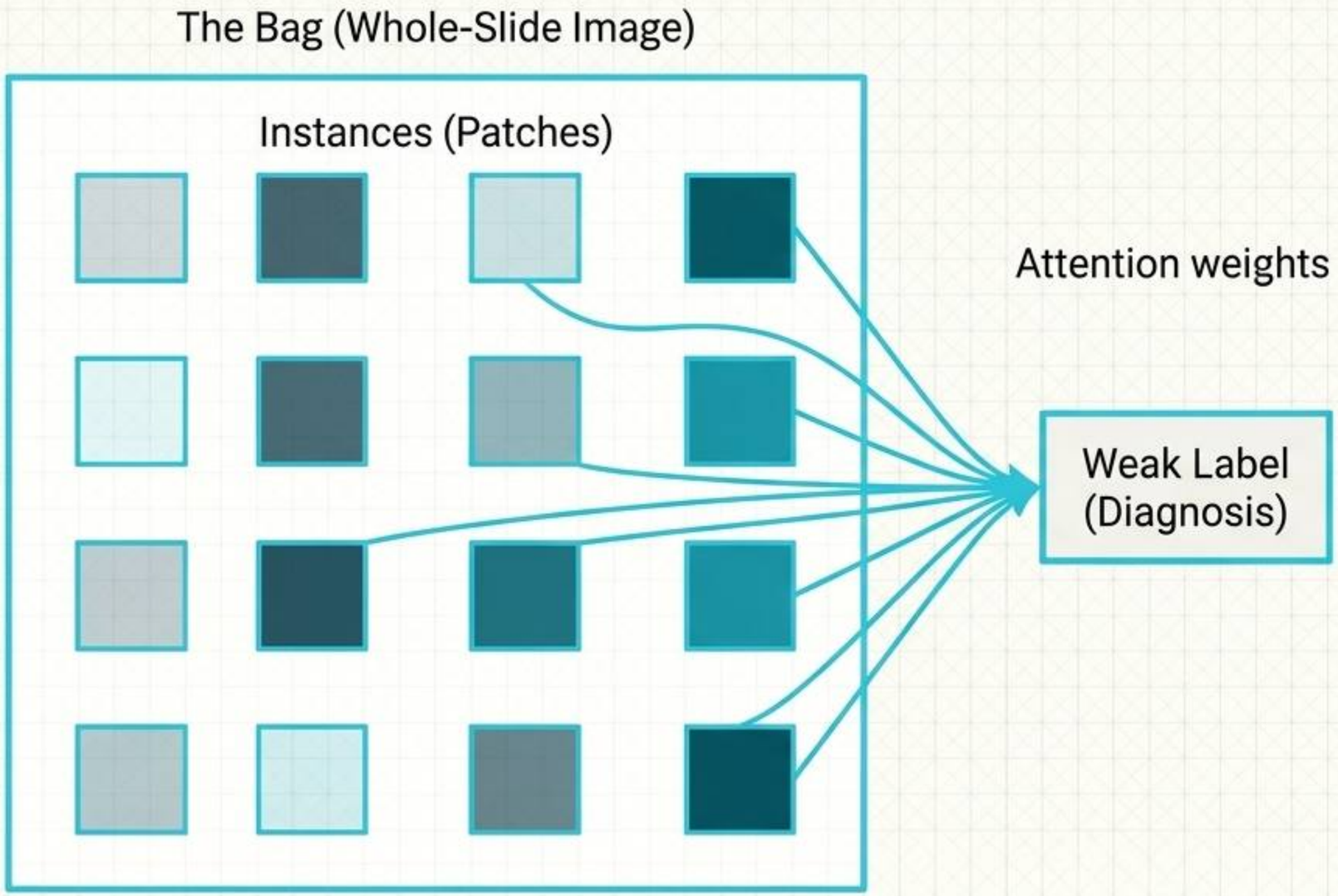
Generated Artifact:
papers/CPath/_SUMMARY.md



NotebookLM

Bridging the Label Gap with Multiple Instance Learning

Generated Artifact:
papers/methods/_SUMMARY.md



Methodological Families Compared

- ABMIL (Attention-based aggregation)
- CLAM-style clustering (Instance-based approach)
- DSMIL-style dual streams (Embedding-based approach)
- Transformer-based methods

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papers/foundation models/_SUMMARY.md

papers/.../models/_SUMMARY.md

Analyzing the Current Frontier of Pathology Foundation Models

Model	Model Family	Modality	Scale / Architecture	Relevance
UNI	Extractor	Unimodal	Large-scale SSL	Baseline Extractor
Virchow (1 & 2)	Extractor	Unimodal	Million-slide scale	Core comparison
RudolfV	Extractor	Unimodal	Self-supervised	Domain-specific evaluation
Prov-GigaPath	Hybrid	Unimodal	Whole-slide scale	Contextual aggregation
PLIP / CONCH	Extractor	Multimodal	Vision-Language	Text-aligned features
CHIEF / THREADS	Aggregator	Multimodal	Transformer-based	End-to-end task analysis

Takeaway:
The literature reveals a distinct shift from task-specific supervised models to self-supervised and multimodal foundation models.

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Transforming Passive Reading into Reusable Knowledge Artifacts

Terminology Stabilization

`docs/glossary.md`

Defines WSI, H&E, biomarkers, and core metric terminology.

Orientation Material

`docs/faq.md`

Supervisor- and reader-facing explanations bridging technical gaps.

Synthesized Outputs

`papers/CPath/_SUMMARY.md`

`_SLIDES.pdf`

`_DEEP_DIVE.m4a`

Reading generated structured assets designed to support future thesis writing, presentations, and rapid onboarding. These are foundational tools, not final thesis chapters.

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Defining the Inherited Baseline and Transitioning to Experimental Design

The Baseline (Past Context)

Original Setup (Weeks 1-2)

Inherited Baseline

code/TCGA-UT/

Collected solely for understanding prior workflows, dataset handling, and visualization outputs. This is inherited prior-employee context, not original code.

The Build (Weeks 3 & 4)

- find and characterize relevant/available datasets
- research low-data/heavy-tail domain and cpath model calibration
- understand what was done in inherited code

Infrastructure and conceptual groundwork complete. Moving from broad orientation to targeted experimental focus.

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